Number of problems follow the book.

2.1 Use algebraic manipulation to prove that x + yz = (x + y) · (x + z). Note that this is the distributive rule, as stated in identity 12b in Section 2.5.

2.1. The proof is as follows:

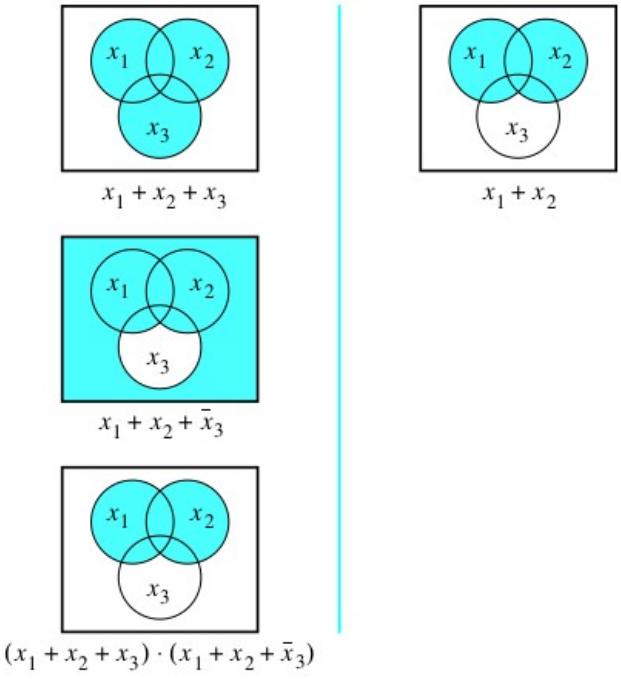
$$\begin{aligned}(x+y)\cdot(x+z) &= xx + xz + xy + yz \\ &= x + xz + xy + yz \\ &= x(1 + z + y) + yz \\ &= x \cdot 1 + yz \\ &= x + yz\end{aligned}$$

2.3 Use algebraic manipulation to prove that xy + yz + xz = xy + xz. Note that this is the consensus property 17a in Section 2.5.

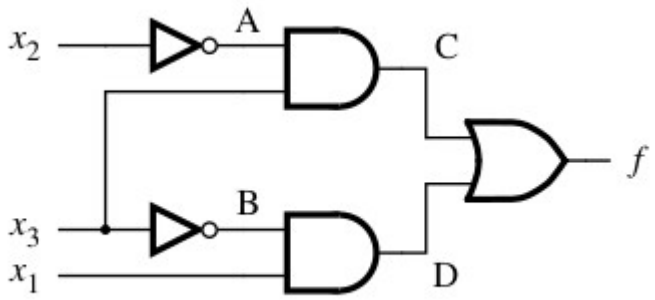
$$\begin{aligned}xy + yz + \overline{x}z &= xy + (x + \overline{x})yz + \overline{x}z \\ &= xy + xyz + \overline{x}yz + \overline{x}z \\ &= xy(1 + z) + \overline{x}(y + 1)z \\ &= xy \cdot 1 + \overline{x} \cdot 1 \cdot z \\ &= xy + \overline{x}z\end{aligned}$$

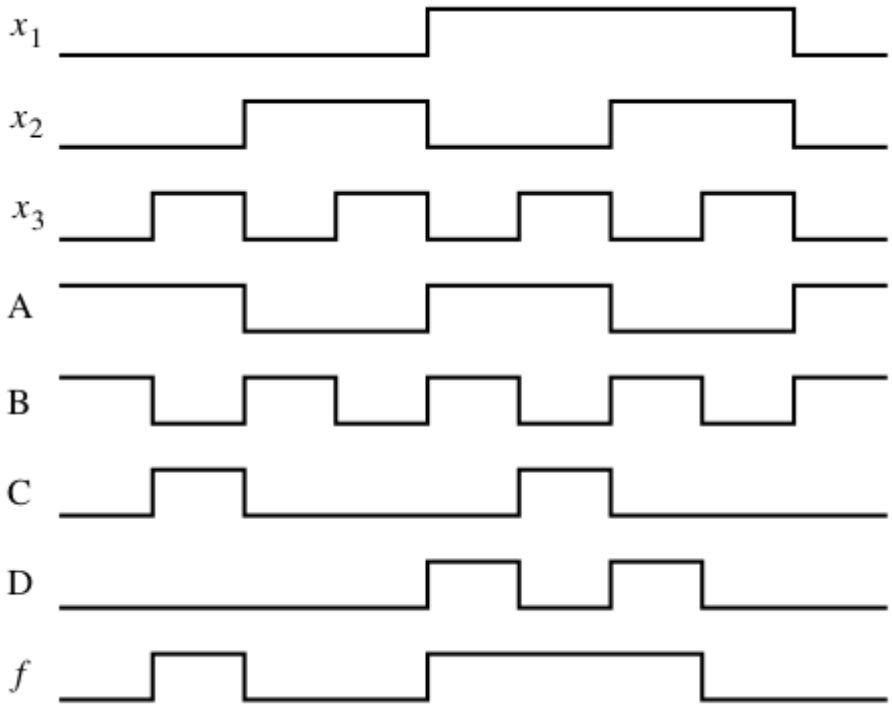
2.6 Use the Venn diagram to prove that
(x 1 + x 2 + x 3) · (x 1 + x 2 + x 3) = x 1 + x 2

2.6. Proof using Venn diagrams:



2.8 Draw a timing diagram for the circuit in Figure 2.24a. Show the waveforms that can be observed on all wires in the circuit.





2.11 Use algebraic manipulation to show that for three input variables x_1 , x_2 , and x_3
#M (0, 1, 2, 3, 4, 5, 6) = x_1 x_2 x_3

$$
\begin{aligned}
f &= (x_1 + x_2 + x_3)(x_1 + x_2 + \overline{x}_3)(x_1 + \overline{x}_2 + x_3)(x_1 + \overline{x}_2 + \overline{x}_3) \cdot \\
&\quad (\overline{x}_1 + x_2 + x_3)(\overline{x}_1 + x_2 + \overline{x}_3)(\overline{x}_1 + \overline{x}_2 + x_3) \\
&= ((x_1 + x_2 + x_3)(x_1 + x_2 + \overline{x}_3))((x_1 + \overline{x}_2 + x_3)(x_1 + \overline{x}_2 + \overline{x}_3)) \cdot \\
&\quad ((\overline{x}_1 + x_2 + x_3)(\overline{x}_1 + x_2 + \overline{x}_3))((\overline{x}_1 + \overline{x}_2 + x_3)(\overline{x}_1 + x_2 + x_3)) \\
&= (x_1 + x_2 + x_3\overline{x}_3)(x_1 + \overline{x}_2 + x_3\overline{x}_3) \cdot \\
&\quad (\overline{x}_1 + x_2 + x_3\overline{x}_3)(\overline{x}_1 + \overline{x}_2 x_2 + x_3) \\
&= (x_1 + x_2)(x_1 + \overline{x}_2)(\overline{x}_1 + x_2)(\overline{x}_1 + x_3)
\end{aligned}
$$

$$
\begin{aligned}
&= (x_1 + x_2\overline{x}_2)(\overline{x}_1 + x_2x_3) \\
&= x_1(\overline{x}_1 + x_2x_3) \\
&= x_1\overline{x}_1 + x_1x_2x_3 \\
&= x_1x_2x_3
\end{aligned}
$$

2.21 Design the simplest sum-of-products circuit that implements the function f (x1 , x2 , x3) = ∑m(1, 3, 4, 6, 7).

$$
\begin{aligned}
f &= \overline{x}_1\overline{x}_2x_3 + \overline{x}_1x_2x_3 + x_1\overline{x}_2\overline{x}_3 + x_1x_2\overline{x}_3 + x_1x_2x_3 \\
&= \overline{x}_1(\overline{x}_2 + x_2)x_3 + x_1(\overline{x}_2 + x_2)\overline{x}_3 + (\overline{x}_1 + x_1)x_2x_3 \\
&= \overline{x}_1x_3 + x_1\overline{x}_3 + x_2x_3
\end{aligned}
$$

Another possibility is

$$
f = \overline{x}_1x_3 + x_1\overline{x}_3 + x_1x_2
$$

2.31 For the timing diagram in Figure P2.3, synthesize the function f (x1 , x2 , x3) in the simplest sum-of-products form.

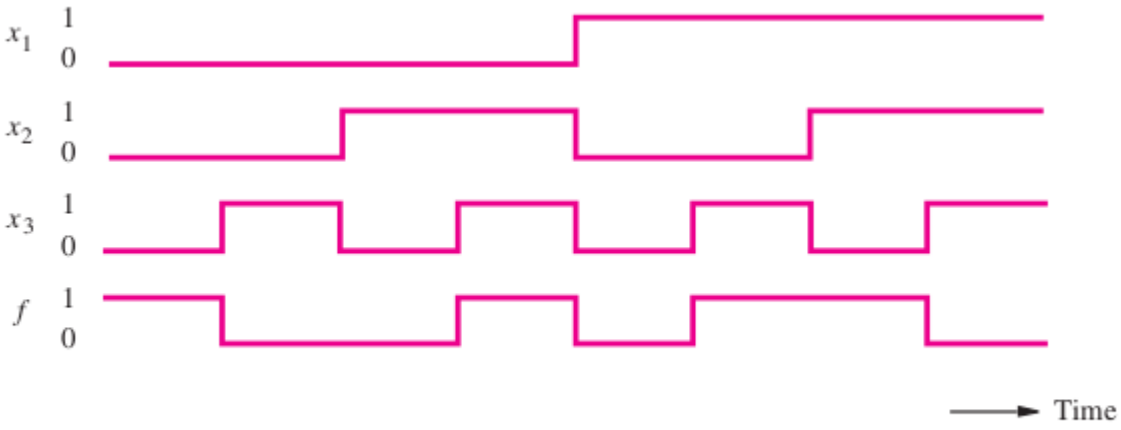


Figure P2.3 A timing diagram representing a logic function.

x_1	x_2	x_3	f
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

The simplest SOP expression is $f = \overline{x}_1\overline{x}_2\overline{x}_3 + \overline{x}_1 x_2 x_3 + x_1\overline{x}_2 x_3 + x_1 x_2\overline{x}_3$.

2.32 For the timing diagram in Figure P2.3, synthesize the function f (x 1 , x 2 , x 3) in the simplest product-of-sums form.

x_1	x_2	x_3	f
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

The simplest POS expression is $f = (x_1 + x_2 + \overline{x}_3)(x_1 + \overline{x}_2 + x_3)(\overline{x}_1 + x_2 + x_3)(\overline{x}_1 + \overline{x}_2 + \overline{x}_3)$.

2.35 Design a circuit with output f and inputs x1 , x0 , y1 , and y0 . Let X = x1 x0 and Y = y1 y0 represent two 2-digit binary numbers. The output f should be 1 if the numbers represented by X and Y are equal. Otherwise, f should be 0.
(a) Show the truth table for f.
(b) Synthesize the simplest possible product-of-sums expression for f.

2.35. (a)

x_1	x_0	y_1	y_0	f
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

(b) The simplest POS expression is $f = (x_1 + \overline{y}_1)(\overline{x}_1 + y_1)(x_0 + \overline{y}_0)(\overline{x}_0 + y_0)$.